2025 International Conference on New Trends in Computational Intelligence (NTCI) | 979-8-3315-5614-3/25/$31.00 ©2025 IEEE | DOI: 10.1109/NTCI67886.2025.11308230

FusionOM: A Dual-Channel LLM–KG Framework for Intelligent Operations and Maintenance Strategy Generation

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***Abstract*—Operations and Maintenance (O&M) strategies in safety-critical systems often rely on manual workflows that are time-consuming and error-prone. Recent advances in Large Language Models (LLMs) and Knowledge Graphs (KGs) enable new opportunities for intelligent, automated O&M decision support. This manuscript proposes FusionOM, a dual-channel retrieval and reasoning framework that integrates semantic Retrieval-Augmented Generation (RAG) with Graph-based RAG (GraphRAG). The framework combines natural language understanding with structured domain knowledge to generate accurate, explainable, and context-aware maintenance strategies.Using 74 civil aviation accident reports as a case study, FusionOM improves cross-report knowledge coverage and reasoning faithfulness, shortens strategy generation from weeks to hours, and outperforms recent RAG/GraphRAG variants in coverage and consistency. The framework enables standardized, reusable reports and offers a scalable foundation for domain-adaptive O&M decision support across industries.**

***Index Terms*—Intelligent Operations and Maintenance (O&M); Large Language Models (LLMs); Knowledge Graphs (KGs); Retrieval-Augmented Generation (RAG)**

I. Introduction

Operations and Maintenance play a critical role in ensuring the safety, reliability, and efficiency of complex systems such as civil aviation, power grids, and manufacturing [1]. Traditional O&M strategy development is heavily dependent on manual analysis of technical documents and expert knowledge, which is both time-consuming and prone to inconsistencies. With the rapid growth of digital data, there is an urgent need for intelligent methods that can transform unstructured information into actionable maintenance strategies [2].

LLMs have recently demonstrated remarkable capabilities in natural language understanding and knowledge synthesis. However, their application in O&M is hindered by two key challenges: (1) hallucination risks when reasoning without grounded knowledge [3], and (2) limited explainability in decision-making processes [4]. KGs, on the other hand, provide structured, explainable knowledge representations but lack the generative capabilities required for comprehensive strategy formulation [5]. Existing works typically employ either LLM-based retrieval-augmented generation (RAG) or KG-based reasoning in isolation, which restricts their performance in practical O&M applications.

To address these challenges, we present FusionOM (Fusion of Large Language Models and Knowledge Graphs for Operations and Maintenance), a dual-channel retrieval framework that integrates semantic RAG with GraphRAG. By integrating the generative capabilities of LLMs with the structured reasoning of KGs, FusionOM enables accurate, explainable, and efficient O&M strategy generation. Using civil aviation accident reports as a case study, we demonstrate that FusionOM substantially improves coverage, interpretability, and efficiency compared with baseline methods.

The key contributions of this paper are summarized as follows:

- *Framework Design* - We propose FusionOM, a dual-channel O&M framework that fuses semantic retrieval with KG-based reasoning for strategy generation.
- *Knowledge Integration* - We design a hybrid retrieval mechanism that balances unstructured semantic understanding with structured domain knowledge.
- *Case Study and Evaluation* - We validate the framework on 74 civil aviation accident reports, achieving a 60× increase in efficiency while maintaining explainability.
- *Scalability and Applicability* - we discuss the potential extension of the framework to other critical domains such as power systems and manufacturing.

The remainder of this paper is organized as follows:

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Section II reviews related work; Section III presents the FusionOM framework; Section IV describes the experimental setup and results; and Section V concludes the paper.

II. Related Work

Research on intelligent O&M decision support has been explored from three major perspectives: LLMs, Retrieval-Augmented Generation (RAG), and Knowledge Graphs (KGs).

LLM-based O&M. Recent advances in LLMs have enabled natural language understanding for maintenance report analysis and fault diagnosis [6]. While these models can generate fluent strategies, they often suffer from hallucinations and lack domain-specific grounding, which limits their reliability in safety-critical contexts [7].

Retrieval-Augmented Generation (RAG). To mitigate hallucinations, RAG integrates document retrieval with LLM reasoning [8]. Prior studies have shown improvements in factual accuracy for technical domains. However, RAG methods typically rely on unstructured semantic retrieval, which does not fully leverage structured domain knowledge.

Knowledge Graphs (KGs). KGs have been applied to model O&M knowledge through graph reasoning, offering explainability and interpretability [9]. Yet, KG-only approaches struggle with coverage and flexibility, since constructing and maintaining large-scale graphs is costly, and they lack generative capabilities.

Limitations. Existing works thus adopt either RAG-only or KG-only methods, leading to trade-offs between coverage, accuracy, and explainability. Few studies attempt to combine LLMs and KGs in a unified framework tailored for O&M strategy generation.

III. Methodology

To address three critical issues in complex system operation and maintenance (O&M)—*insufficient knowledge coverage*, *low response efficiency*, and *difficulty in systematically consolidating investigation results*—we propose an intelligent O&M scheme generation framework, FusionOM, that integrates LLMs with KGs. The core idea is a *dual-channel* mechanism, combining semantic retrieval with graph-based reasoning to perform parallel evidence gathering and constrained generation. The framework aims to produce O&M schemes that are accurate, consistent and explainable while supporting the accumulation of iterative knowledge.

As illustrated in Fig.1, FusionOM comprises four modules: (i) knowledge base construction; (ii) dual-channel retrieval; (iii) prompt construction and scheme generation; and (iv) feedback updating. In the following sections, we describe each module in detail.

A. Knowledge Graph Construction

Knowledge base construction forms the foundation of the proposed *FusionOM* framework, supporting dual-channel retrieval and reasoning. It consists of two complementary levels: semantic representation and structured graph modeling.

At the **semantic level**, the original documents $D = \{d_i\}_{i=1}^{N}$ are segmented into chunks $C = \{c_j\}_{j=1}^{M}$, each encoded as a dense vector $f_\theta(c_j)$ with metadata $m_j = \{\text{source}, \text{timestamp}, \text{scene}\}$ for provenance tracking.

At the **structured level**, an LLM fine-tuned on O&M corpora using LoRA is employed to extract entities and relations, forming a knowledge graph $G = (V, E)$. The node set V represents O&M-related entities such as equipment, components, fault modes, and corrective actions, while the edge set E denotes semantic or causal relations expressed as triples $e = (v_s, r, v_t)$. All embeddings preserve provenance metadata to ensure auditability and traceability.

To **unify the two levels**, we define a mapping function $\text{Link}(\cdot)$ that connects graph entities and relations to their supporting semantic chunks. All embeddings (chunks, nodes, and relations) are stored in a unified vector index with type and metadata annotations.

B. Dual-Channel Retrieval

The dual-channel retrieval module consists of semantic retrieval and knowledge graph reasoning. In the semantic retrieval channel, the system selects the Top-K_c chunks most relevant to the query q based on cosine similarity $s(x,y) = \cos(x,y)$, yielding the candidate set:

$$R_{\text{text}}(q) = \text{Top-}K_c \; s(q, c_j) \mid c_j \in C. \tag{1}$$

In the knowledge graph reasoning channel, the system first performs entity-level coarse matching:

$$R_V(q) = \text{Top-}K_v \; s(q, v) \mid v \in V, \tag{2}$$

where K_c and K_v are the numbers of selected chunks and entities, respectively. Based on $R_V(q)$, a query-related subgraph is then constructed according to the reasoning strategy ρ:

$$G_q = \text{Closure}_\rho(R_V(q), E), \tag{3}$$

where ρ may impose constraints such as relation priority, depth limit L, or relation-type filtering. $\text{Closure}_\rho(\cdot)$ denotes the recursive subgraph expansion starting from $R_V(q)$ and adding adjacent nodes and edges from E. From this subgraph, the system collects relevant textual evidence:

$$C_{\text{graph}}(q) = \bigcup_{v \in G_q} \text{Link}(v). \tag{4}$$

Finally, the evidence from both channels is merged into a unified set:

$$S(q) = R_{\text{text}}(q) \cup C_{\text{graph}}(q). \tag{5}$$

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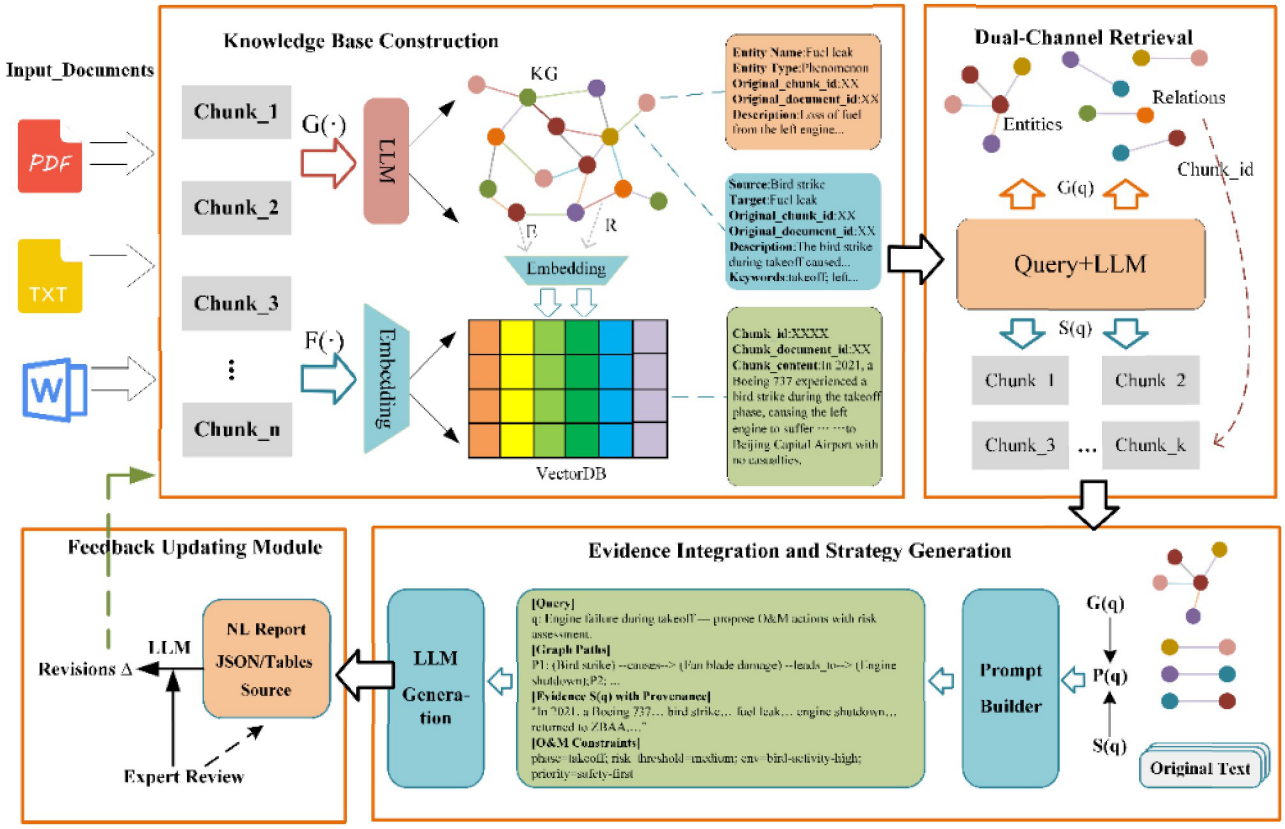


Fig. 1. Overall architecture of the proposed *FusionOM* framework. The system consists of four core modules: knowledge base construction, dual-channel retrieval, prompt construction and strategy generation, and feedback updating. Semantic retrieval (RAG) and graph reasoning (GraphRAG) work in parallel to collect evidence, while the LLM generates standardized and explainable O&M strategies under KG constraints.

This dual-channel design enables the system to capture explicit information through semantic similarity, while leveraging graph reasoning to uncover implicit causal chains and upstream–downstream dependencies. As a result, semantic retrieval is complemented by indirect evidence, thereby improving evidence recall and interpretability of reasoning.

C. Evidence Integration and Strategy Generation

After semantic retrieval and graph reasoning, the system constructs a structured prompt from the evidence set $S(q)$ and the subgraph G_q:

$$P(q) = \text{Format}(q, G_q, S(q)), \tag{6}$$

where $\text{Format}(\cdot)$ serializes G_q into triples or paths, aligns $S(q)$ with metadata (e.g., source, timestamp), and incorporates O&M constraints such as risk level and execution priority.

A fine-tuned large language model LLM_ϕ, trained on O&M corpora using LoRA, then generates standardized maintenance strategies:

$$\hat{A}(q) = \text{LLM}_\phi(P(q)). \tag{7}$$

This module outputs both natural language solutions and structured formats (e.g., JSON, tables) including event descriptions, reasoning paths, recommended actions, and risk assessments. Such outputs allow strategies to be directly embedded into O&M plans for system integration and automation. LoRA fine-tuning enhances domain understanding and controllability, while few-shot examples and template prompts improve consistency, ensuring practical applicability in real-world O&M scenarios.

D. Feedback Updating Module

After solution generation, domain experts review the output $\hat{A}(q)$ and provide revision feedback Δ. This feedback is parsed into knowledge increments:

$$F(\Delta) \rightarrow \{\Delta V, \Delta E, \Delta C\}, \tag{8}$$

where ΔV denotes added or modified entities, ΔE represents updated relations, and ΔC indicates supplemental textual evidence. These increments update the knowledge graph G and the chunk set C, with synchronized refresh of their embeddings and metadata.

Through this process, the system achieves closed-loop knowledge evolution: generated solutions are validated and refined by expert review, and revisions are injected back

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into the knowledge base. Consequently, the knowledge graph and semantic repository are continuously enriched, ensuring dynamic adaptation and long-term accumulation of O&M expertise.

E. Experimental Application

Inference Workflow. During inference, FusionOM executes a closed-loop process as shown in Algorithm 1.

Algorithm 1: FusionOM: Closed-Loop O&M Strategy Generation

Input: Query q; Knowledge graph $G = (V, E)$; Repository C
Output: Validated strategy $\hat{A}(q)$
Step 1: Retrieval ;
$R_{\text{text}} \leftarrow$ Top-K_c chunks from C; $R_V \leftarrow$ Top-K_v entities from V; $G_q \leftarrow \text{Closure}_\rho(R_V, E)$;
$S(q) \leftarrow R_{\text{text}} \cup \bigcup_{v \in G_q} \text{Link}(v)$;
Step 2: Prompting ;
$P(q) \leftarrow \text{Format}(q, G_q, S(q))$;
Step 3: Generation ;
$\hat{A}(q) \leftarrow \text{LLM}_\phi(P(q))$;
Step 4: Review & Update ;
$\Delta \leftarrow$ expert revisions on $\hat{A}(q)$; Update (G, C) with $F(\Delta)$;
return $\hat{A}(q)$;

Advantages. By integrating semantic retrieval with structured graph reasoning, the framework improves both accuracy and interpretability of maintenance strategies. Provenance tracing and temporal control ensure that generated results remain verifiable and timely, while the closed-loop update mechanism continuously enriches domain knowledge, completing the decision chain from data to knowledge to action.

IV. Experimental Analysis

A. Dataset Description

To evaluate the effectiveness of the proposed FusionOM framework, we collected and processed 74 civil aviation accident investigation reports. From each report, factual events were abstracted into *component + failure phenomenon* pairs (e.g., "flap jam," "fuel tank leakage"). The dataset spans four major families: *structure*, *flight control*, *fuel system*, and *environmental conditions*, providing a representative benchmark for evaluating knowledge coverage, efficiency, and consistency.

B. Experimental Setup

The FusionOM framework was configured as follows:

- A domain-adapted LLM (fine-tuned with LoRA) was used for entity/relation extraction and strategy generation.
- Dual-channel retrieval was applied: semantic retrieval (RAG) for explicit evidence, and graph reasoning (GraphRAG) for implicit causal chains.
- Retrieved evidence and reasoning subgraphs were converted into structured prompts with domain constraints.
- Expert-in-the-loop feedback enabled iterative enrichment of the knowledge base.

Two main evaluation aspects were considered:

- **Knowledge Coverage Analysis** — to quantify how well FusionOM reconstructs cross-report causal chains by supplementing primary, secondary, and incidental information.
- **O&M Efficiency and Baseline Comparison** — to measure the acceleration achieved compared with manual workflows and to benchmark FusionOM against recent RAG/GraphRAG variants in terms of consistency and explainability.

Experimental Environment. All experiments were conducted on a workstation equipped with four NVIDIA L20 GPUs, using LoRA for lightweight fine-tuning of the base 32B LLM.

1) Coverage and Efficiency Analysis: We first evaluate the ability of FusionOM to enhance knowledge coverage. Coverage is defined as:

$$Coverage = \frac{M_{\text{prim-doc}}}{M}, \tag{9}$$

where M is the total number of facts in a component category, and $M_{\text{prim-doc}}$ denotes those appearing in primary-cause documents. Accordingly, $1 - Coverage$ reflects the knowledge gap.

As shown in Table I, coverage for the fuel system and environmental conditions reaches about 73%–75%, flight control about 68%, while structural components show the lowest coverage (61%), indicating substantial omissions when only primary-cause documents are considered. Fig. 2 further illustrates inter-category variations, where some components (e.g., *wing*, *winglet*) remain below 60%. FusionOM effectively supplements secondary and incidental evidence, thereby improving cross-report causal chain reconstruction.

TABLE I
Knowledge Coverage and Gaps Across Causal Families

Family	Total M	Primary M_p	Coverage (%)	Gap (%)
Structure	547	333	60.88	39.12
Flight Control	458	312	68.12	31.88
Fuel System	239	178	74.48	25.52
Environmental	231	169	73.16	26.84

We also measure the efficiency of strategy generation. As shown in Table II, the conventional manual workflow required **720–960 hours** (30–40 days) for end-to-end processing, while FusionOM reduced the cycle to less than **12 hours**, yielding an overall **60× speedup** with higher logical consistency and stability.

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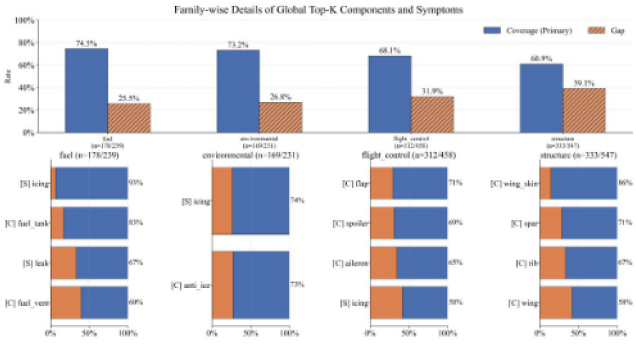


Fig. 2. Coverage vs. gaps across families (top) and representative components (bottom). Blue=coverage, orange=gap; K=4.

TABLE II
EFFICIENCY COMPARISON: MANUAL PROCESS VS. FUSIONOM

Workflow Stage	Manual Process	FusionOM
Evidence collection and documentation	48–72 h	— (integrated in KB)
Manual retrieval and literature review	120–168 h	1–2 h
Key information extraction and note-taking	168–240 h	Automated by LLM
Causal chain reconstruction and cross-case comparison	120–168 h	Minutes (via KG reasoning)
Plan drafting and collective review	240–360 h	2–3 h (structured output)
Total cycle time	**720–960 h**	**< 12 h**

2) Comparative Evaluation: To further evaluate the effectiveness of FusionOM, we compare it with representative RAG and GraphRAG variants proposed in the past 1–2 years, including **Semantic RAG, GraphRAG, E^{2}GraphRAG [10], LightRAG [11], and FG-RAG [12]**. The evaluation is conducted using five criteria: *Cov* (domain knowledge coverage), *F* (faithfulness of generated content), *AR* (answer relevance to queries), *CR* (context recall in retrieval), and *Cons* (logical consistency across repeated trials).

As shown in Table III, recent methods achieve varying degrees of improvement in grounding and relevance but remain limited in coverage, recall, and stability. FusionOM demonstrates the highest Cov (>80%), F (88%), and Cons (>80%), while also maintaining superior AR and CR, thereby establishing its advantage over contemporary approaches.

TABLE III
COMPARISON WITH RECENT REPRESENTATIVE RAG/GRAPHRAG METHODS

Method	Cov	F	AR	CR	Cons
Semantic RAG (2023)	0.55	0.70	0.58	0.55	0.50
LightRAG (2024)	0.57	0.62	0.60	0.56	0.52
GraphRAG (2024)	0.62	0.78	0.65	0.62	0.60
E^{2}GraphRAG (2025)	0.66	0.72	0.69	0.65	0.67
FG-RAG (2025)	0.70	0.78	0.72	0.68	0.70
FusionOM (Ours)	**0.83**	**0.88**	**0.84**	**0.76**	**0.81**

V. CONCLUSIONS

This paper proposes FusionOM, a dual-channel framework that integrates semantic RAG with Knowledge GraphRAG for intelligent O&M strategy generation. By combining the generative strengths of LLMs with the structured reasoning of KGs, FusionOM addresses key challenges of hallucination, limited explainability, and incomplete coverage in existing approaches. Experiments on 74 civil aviation accident investigation reports demonstrate that FusionOM accelerates strategy generation by over **60**× compared with manual workflows, while also outperforming recent RAG and GraphRAG baselines in knowledge coverage, faithfulness, and consistency. Beyond these improvements, FusionOM enables standardized, reusable, and explainable O&M strategies, reducing reliance on manual expertise and supporting more timely and reliable decision-making.

ACKNOWLEDGMENT

This work was supported in part by the National Natural Science Foundation of China under Grant No. 42371476, Fundamental Research Funds for the Central Universities (buctrc202132).

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